# Recursive Portals and Prediction on Unseen Data

Executed research milestone | 3 November 2026 | Negative / conditional outcome

Concept originator: Dan A. Merluse. AI-assisted experiment design, code, calculations and reporting. Research-stage evaluation, not a verified native SRDE deployment.

## Executive decision

The current portal-inspired predictor did not outperform strong ordinary graph methods on a public genuinely unseen-entity benchmark. It essentially tied a provenance-aware graph-consensus scorer and trailed a simple 2-3 hop Katz-style baseline. Synthetic evidence shows that state and source lineage can help when the data-generating process truly depends on them, but ordinary graph learning can use the same signal.

| Experiment | Portal AUC | Best conventional AUC | Conclusion |
|---|---|---|---|
| Inductive real graph (205 facts) | 0.6083 | Katz 0.6638 | No portal advantage |
| Synthetic unseen worlds (9,000 cases) | 0.6903 | Graph HGB 0.6937 | No portal advantage |
| Synthetic shifted confirmation (6,000) | 0.6247 | Graph HGB 0.6266 | No stable advantage |

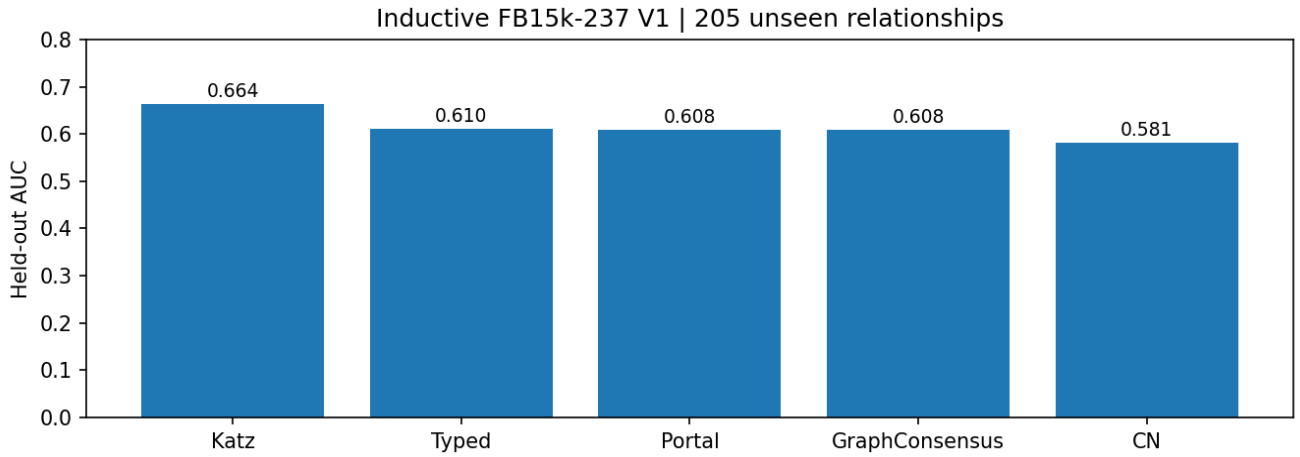
## Prior SRDE evidence respected

The Nov 2 portal hypothesis specifies conditional transitions, position/orientation/state/history, affective coordinates, source provenance, decaying evidence and worker consensus. Its synthetic 100-level example warns that random long chains can connect nearly anything; it is not pre-event forecasting proof. The Oct 12 local-model working paper found 55/56 correct evidence-assisted answers versus static graph retrieval 48/56, but a later intent-aware graph holdout tied SRDE at 56/56. The E30 local report recorded SRDE 38/80 versus graph-state 43/80, and a confirmation tie at 18/40.

## Experiment A | Genuine unseen-entity relational prediction

GraIL FB15k-237 V1: 4,245 training triples among 1,594 entities; a separate inference graph has 1,993 triples among 1,093 entirely different entities (zero overlap); 206 validation and 205 held-out test triples. A deterministic 20-negative candidate task was created for each target relation with known positives excluded from evaluation negative sampling. This is unseen-entity generalization, not true chronological event forecasting.

A relation-aware path-rule learner searched 2-3 hop paths, masked each target training edge, fit 2,378 association patterns using training positives and 2 corrupt negatives per positive. Eight scoring variants used the same enumerated path evidence. The portal-inspired variant grouped witness paths by first intermediary and stopped at depth two when two independent positive branches were observed; otherwise it included depth-three paths. This is a structural portal approximation, not the full SRDE algorithm.




| Method | Test AUC | Test MRR | Assessment |
|---|---|---|---|
| Katz 2-3 hop | 0.6638 | 0.3556 | Best of tested |
| Typed graph | 0.6101 | 0.3129 | Marginally > portal AUC |
| Portal prototype | 0.6083 | 0.3142 | Near consensus tie |
| Graph consensus | 0.6080 | 0.3118 | Near portal tie |
| Common neighbors | 0.5810 | 0.2774 | Weaker topology |

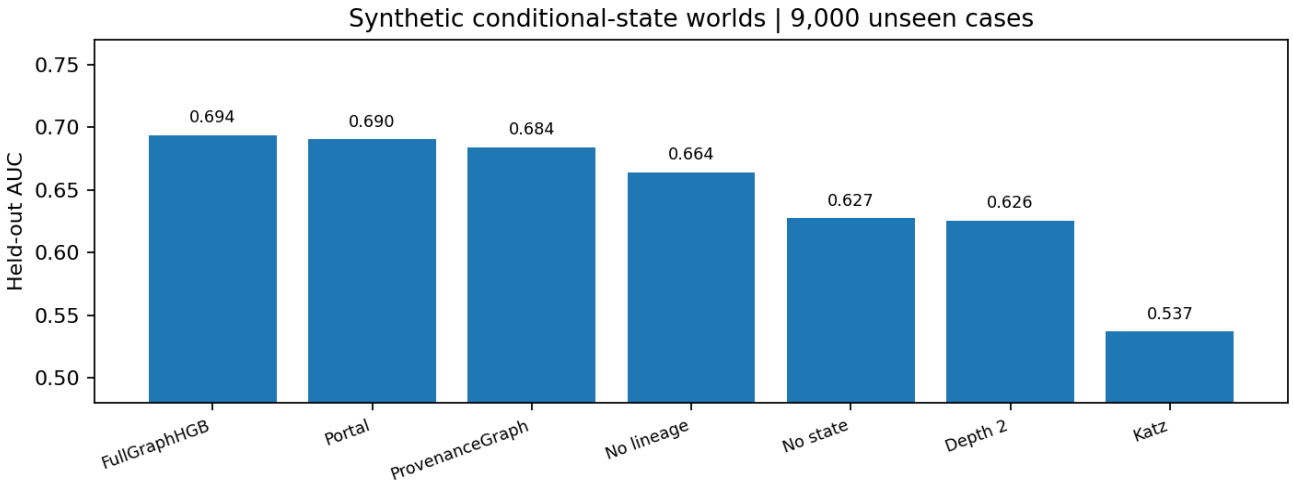
### *Uncertainty and coverage*

On 205 paired test positives, Portal - Katz AUC = -0.05549, with 95% 2,000-resample query-bootstrap interval [-0.08780, -0.02439]. Portal - conventional graph consensus = +0.000244 with interval [0.000000, 0.000732]: this tiny difference is effectively one affected query and not a meaningful improvement. Only 49/205 held-out positives had any positive learned rule path. All candidate sets contained 20 negatives; none of the graph traversals reached the 12,000-expansion cap.

Critical caveats: ranking uses sampled negatives rather than official all-entity filtered metrics; one V1 dataset and seed; no published neural GNN/GraIL/NBFNet/SEAL implementation; no time-indexed future data, learned affect, or full recursive worker behavior. Results cannot establish calibrated event probabilities or robust deployment economics.

## Experiment B | Conditional-state and provenance stress test

Synthetic worlds: 80 training (12,000 examples), 20 validation (3,000), 60 new test (9,000) and 40 shifted confirmation (6,000). Each pair receives independent source-level signs, context tags, orientation, reliability, age/decay and 2-3 hop evidence; duplicated paths retain a common provenance ID. The generator makes labels probabilistically depend on independent, context-active source support. This deliberately matches the candidate mechanism, so it is a diagnostic feasibility test rather than independent semantic proof.




| Method | Unseen AUC | Shift AUC | Interpretation |
|---|---|---|---|
| Full graph HGB | 0.6937 | 0.6266 | Conventional flexible learner |
| Full graph logistic | 0.6913 | 0.6266 | Conventional same-input model |
| Portal | 0.6903 | 0.6247 | No superior prediction |
| Provenance graph | 0.6839 | 0.6231 | Lineage useful |
| Portal, no lineage | 0.6640 | 0.6187 | Duplicate-vote error |
| Portal, no state | 0.6273 | 0.5656 | Context matters |
| Portal, 2 hops | 0.6256 | 0.5766 | Depth helps this generator |
| Katz | 0.5370 | 0.5261 | Topology insufficient here |

Portal - HGB AUC is -0.00337 pooled on unseen synthetic cases. Resampling entire test worlds 500 times yielded a 95% interval for world-mean AUC difference [-0.00566, -0.00081]. In shifted confirmation, the conventional full graph learner also edged portal. Unlike Experiment A, this is within a designer-authored generator which directly specifies the role of state and provenance.

### *Negative controls*

Shuffled synthetic test labels yielded portal AUC 0.4952 (near chance). In 200 entirely random 100-layer graphs, average reachable endpoints = 9.795 of 12, illustrating why a long portal path alone is not evidential. Integrity assertions passed for separated seeds/worlds, exact case counts and bounded path observations.

## What is inferable - and what is not

Supported for the tested scope: ordinary 2-3 hop graph topology beat the structural portal prototype on unseen-entity relational ranking. Structural path and relation rule learning is not automatically better than simple connectivity.

Conditional: context gating, decayed reliability and source independence can improve predictions when the unknown target really depends on such variables; conventional models can use the same evidence.

Not supported: that long chains prove meaningful or causal links; that affective states are faithfully encoded by synthetic tags; that recursive portals forecast future events, unlock universal semantic geometry or outperform trained modern

graph neural networks.

New research priority: explicitly distinguish relation existence from causal direction, false-assertion cost from abstention, independent source IDs from multiple copies, and pre-event evidence timestamps from post-event leakage.

## Next rigorous experiment / gate

Expand to independent inductive graph families (FB15k-237 V2-V4 and WN18RR where suitable), hard HeaRT-style negative sampling, and a chronological prediction dataset (candidate ICEWS/GDELT). Run official evaluation protocols and modern SEAL/GraIL/NBFNet baselines, preserve all methods within equal data access and report full compute. Train and choose hyperparameters on training/validation only; freeze code before testing; preregister success criteria, then confirm on held-out shifted datasets and repeat across at least five seeds. Add adversarial path copying, correlated upstream sources, random links, orientation/state mislabeling and ablation of recursive depth.

Promote SRDE only if it shows a reproducible positive uplift over the strongest conventional matched comparator, confidence intervals exclude zero across multiple independent data sources, and calibration, false positives, abstention and compute costs remain acceptable. No such advancement is supported yet.

## Provenance and supplied artifacts

Public benchmark read directly through GitHub-connected file tools. Four source path names and Git object SHA1 hashes are preserved in real_inductive_results.json. A Node reproduction script is included and syntax/synthetic-fixture smoke-tested; no second full real-data Node rerun occurred in the isolated container because the external raw GitHub endpoint was inaccessible. The real observed results came from the independent GitHub-connected evaluator, not that smoke fixture.

Synthetic run executed locally using Python / NumPy / scikit-learn. Frozen protocol, source code, raw held-out case CSV, all per-method metrics and JSON receipts are included. The PDF and Markdown are summaries and do not replace the raw receipts.

## Research references

SRDE Portal-Based Hierarchical State Spaces for Pre-Event Inference, v0.1, Nov 2 2026 (source in user Library).

FACO-SRDE Working Paper, Oct 12 2026 (source in user Library).

FACO-SRDE E30 Local Results Report, Oct 16 2026 (Drive file ID 1rWVGiB4-ngjEnNDwP-apC6U3U_kfb1Kt).

GraIL original data and code: https://github.com/kkteru/grail

Li et al., Evaluating Graph Neural Networks for Link Prediction: Current Pitfalls and New Benchmarking: https://arxiv.org/abs/2306.10453

Zhang et al., Revisiting Graph Neural Networks for Link Prediction: https://arxiv.org/abs/2010.16103